PHY-112 | Principles of Physics-2

Review | Final talk

Department of Mathematics & Natural Sciences

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Let's talk about the Final

Main Syllabus
On paper

- Potentials of Continuous Sources | Chapter 24 (Resnick, Halliday) | Chapter 23 (Young, Freedman)
- Capacitors and Capacitance | Chapter 25 (Resnick, Halliday) | Chapter 24 (Young, Freedman)
- Electric Currents and Resistance | Chapter 26 (Resnick, Halliday) | Chapter 25 (Young, Freedman)
- Electric (DC) Circuits and RC Circuits | Chapter 27 (Resnick, Halliday) | Chapter 26 (Young, Freedman)
- Magnetic Forces | Chapter 28 (Resnick, Halliday) | Chapter 27 (Young, Freedman)
- Magnetic Field Sources | Chapter 29 (Resnick, Halliday) | Chapter 28 (Young, Freedman)
- Electromagnetic Induction | Chapter 30 (Resnick, Halliday) | Chapter 29 (Young, Freedman)

What to expect on the Final exam?

Tentative format?
Might be subject to change

- The final exam will be a comprehensive one
- 4 questions in total with sub-questions. Answer only 3.
- No dedicated Mid question will be in it, but...
- You need only to review the stuff that shows up or relates to the final syllabus
- Mixture between numerical application and concept-based understanding
- Brute force memorization will not be rewarded

How to deal with the Mid syllabus?

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Pair up the topics

This is just a rough estimation

Aspect	Final Syllabus	Mid Syllabus
Potential applicaiton	Uniform, Non uniform cases (Chapter 24-3, 24-5)	Electric Circuits & Capacitors (Chapter 27-1, 27-2 & 25-1)
Potentials & Capacitors	Uniform, Non uniform potential sources (Chapter 24-5)	Parallel-plate, Cylindrical, Spherical capacitors (Chapter 25-2)
Potentials & Electric Circuits	Point source, Equipotential surfaces (Chapter 24-3)	Drift motion, Ohm's Law (Chapter 26-1, 26-4)
Field Vectors & Field Lines	Diverging/Converging field lines from source charges (Chapter 22-1)	Circular, continuous loops centered around the source currents (Chapter 29-1)
Force vectors and Directions	Center-to-center, attractive or repulsive (Chapter 21-1)	Perpendicular to the direction of current/moving charges (Chapter 28-6)
Motion of Charged Particles	Parabolic in nature until parallel (Chapter 22-6)	Cyclotron/Uniform Circular & Helical in nature (Chapter 28-4)
Dipole	Electric forces, torques, & potential energy storage in electric dipoles (Chapter 22-7)	Magnetic forces, torques, & potential energy storage in magnetic dipoles (Chapter 28-8)
Dipole Moment Vector	Electric dipole Moment and alignment with *E*-fields (Chapter 22-7)	Magnetic dipole Moment and alignment with *B*-fields (Chapter 28-8)
Gauss's Law	Electric Field has source charges (Chapter 23-2)	Magnetic fields have no source charges (Chapter 29)
Work Done	Can do work due to conservative forces (Chapter 24-1)	Cannot do work due to nonconservative forces (Chapter 28-4)
Coulomb *E*-fields vs Induced *E*-fields	From source electric charges (Chapter 22-1)	From time-varying *B*-fields (Chapter 30-3)

How to prepare?

Mix and Merge

It's okay if You cannot finish the syllabus

- Review the basics: Use class notes, lecture notes, books, lecture videos
- Practice exercise problems: Use books, or the exercise problem playlist
- Review the list on the last page and cover them in parallel
- Review the graphical representations of important equations. It is a fan favorite
- Focus strongly on the earlier chapters
- Latter chapters are an easy steal, though
- Learn to use units of quantities in calculation

Good luck, everyone!